

1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

بولین الجبرا جارج بول (George Boole) نے بنایا۔

CS XI

Chapter 1: Computer Systems

Lecture 03 of 12

1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

LEARN IT

Boolean algebra was developed by the English mathematician **George Boole**. It is a kind of algebra that deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming — every `if` condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

OR rules	AND rules	NOT rules
$A + 0 = A$	$A \cdot 1 = A$	$(A')' = A$
$A + 1 = 1$	$A \cdot 0 = 0$	$A + A' = 1$
$A + A = A$	$A \cdot A = A$	$A \cdot A' = 0$

WORKED EXAMPLE

Using a law: Simplify $Y = A'B + AB$.

Step 1: Both terms contain B, so take B out: $Y = B(A' + A)$.

Step 2: $A' + A = 1$ (a variable OR its opposite is always 1).

Step 3: $Y = B \cdot 1 = B$.

COMMON MISTAKE

In Boolean algebra "+" means OR, not addition. So $1 + 1 = 1$ (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

اردو میں سمجھیں

بولین الجبرا جارج بول (George Boole) نے بنایا۔ اس میں صرف دو قیمتیں ہوتی ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریشنز AND، OR اور NOT ہیں، اور ان کے ملاپ سے ایکسپریشن (Expression) بنتا ہے۔

CHECK YOURSELF

1. Who developed Boolean algebra?
2. In Boolean algebra, what is $1 + 1$?

Answers: 1. George Boole. 2. 1 (because + means OR).

LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. **AND operation ($\cdot$)** — gives output 1 **only when ALL inputs are 1**. Expression: $Y = A \cdot B$ (often written just AB).
2. **OR operation (+)** — gives output 1 **when ANY input is 1**. Expression: $Y = A + B$.
3. **NOT operation (' or $\bar{}$)** — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression: $Y = A'$ (also written as A with a bar on top, $\bar{A}$).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

WORKED EXAMPLE

Evaluate each expression for $A = 1, B = 0$.

- (a) $A \cdot B = 1 \cdot 0 = 0$ (not all inputs are 1).
- (b) $A + B = 1 + 0 = 1$ (at least one input is 1).
- (c) $B' = 0' = 1$ (NOT flips 0 to 1).
- (d) $A \cdot B' = 1 \cdot 1 = 1$.

REMEMBER IT

AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

اردو میں سمجھیں

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ). OR آپریشن تب 1 دیتا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دو دروازوں والا کمرہ). NOT آپریشن ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے اسے یونری (Unary) کہتے ہیں۔

CHECK YOURSELF

1. What symbol is used for the OR operation?
2. Why is NOT called a unary operation?
3. Find $A + B'$ when $A = 0, B = 1$.

Answers: 1. "+". 2. Because it works on only one operand (one input). 3. $B' = 0$, so $0 + 0 = 0$.

LEARN IT

A **truth table** shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the "complete record" of how the circuit behaves.

How to create a truth table:

1. Count the number of inputs (n).
2. Calculate total rows = 2^n .
3. List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
4. Calculate the output for each combination.

2-input example ($n = 2$, rows = 4)

A	B	Combination
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3-input example ($n = 3$, rows = 8)

A	B	C	Decimal
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0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

REMEMBER IT

Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every 2^{n-1} rows.

WORKED EXAMPLE

Make the truth table of a 3-input AND: $Y = A \cdot B \cdot C$.

Step 1: $n = 3$, so rows = $2^3 = 8$.

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 111 gives 1.

A	B	C	$Y = A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

COMMON MISTAKE

Do not list rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows = 2^n , not $2 \times n$: 3 inputs give 8 rows, not 6.

اردو میں سمجھیں

ٹرو تھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ n ہوں تو قطاریں (Rows) 2 کی طاقت n ہوں گی؛ دو ان پٹ کے لیے 4 اور تین ان پٹ کے لیے 8 قطاریں۔ ان پٹ ہمیشہ بائری ترتیب میں لکھیں۔

CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?
2. In a 3-input table, what is the 6th row (counting from row 1 = 000)?

Answers: 1. $2^4 = 16$. 2. 101.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Boolean algebra uses:

- a. TRUE(1) and FALSE(0) only
- b. All real numbers
- c. ASCII letters
- d. IPv4

MCQ 2. AND is 1 only when:

- a. Every input is 1
- b. Any input is 1
- c. Inputs differ
- d. There is no input

MCQ 3. OR is 1 when:

- a. Any input is 1
- b. All inputs are 0
- c. Inputs are equal
- d. NOT is applied twice

MCQ 4. NOT of 1 is:

- a. 0
- b. 1
- c. 2
- d. A

MCQ 5. Rows in an n-input truth table:

- a. 2^n
- b. n
- c. n^2
- d. $n+1$

MCQ 6. Two-input truth table has:

- a. 4 rows
- b. 2 rows
- c. 8 rows
- d. 16 rows

Extra short questions

1. Who developed Boolean algebra and what two values does it use?
2. Write $Y = A \cdot B$ and $Y = A + B$ in words.
3. Why is NOT called unary?
4. List the four steps to build a truth table.

Extra long question

1. Write complete truth tables for 2-input AND, OR and 1-input NOT. Add one real-life lock/door/switch example each.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. George Boole. TRUE(1) and FALSE(0).

2. AND: output 1 only if all inputs are 1. OR: output 1 if any input is 1.
3. It has only one operand: it inverts a single input.
4. Count inputs n ; rows = 2^n ; list combinations in binary order; compute each output.

Long

1. AND 0001 on AB=11 (two locks). OR 0111 (two doors). NOT 10 / 01 (light switch). Show all four/two rows.

Next lecture: 1.1.7 Basic Logic Gates (AND, OR, NOT)